PRD-016 — Predicted SAT Score Engine

Status: Ready for Implementation
Version: 1.0
Dependencies: PRD-004 — Student Home Dashboard; PRD-008 — Progress; PRD-012 — Diagnostic Assessment; PRD-014 — Adaptive Recommendation Engine

---

1. Purpose

The Predicted SAT Score Engine estimates a student's likely total SAT score range based on their PrepHub performance.

The engine must:

- Generate the first prediction immediately after the diagnostic.
- Display predictions as ranges rather than exact scores.
- Update the prediction after every completed adaptive practice set.
- Use a diagnostic-specific calculation for the initial prediction.
- Use the student's seven current Ability Scores for later predictions.
- Track approximate improvement since the diagnostic.
- Show progress toward the next score range.
- Preserve historical predictions for the Progress page.

The engine predicts only the **total SAT score range**.

It does not display predicted Reading and Writing or Math section scores.

---

2. Displayed Score Ranges

PrepHub must use these fifteen total SAT ranges:

Range index	Predicted SAT range
15	1530–1600
14	1450–1520

13	1370–1440
12	1290–1360
11	1210–1280
10	1130–1200
9	1050–1120
8	970–1040
7	890–960
6	810–880
5	730–800
4	650–720
3	570–640
2	490–560
1	400–480

The student must never be shown a single exact predicted score.

The engine may maintain internal decimal values for calculations, but only the corresponding range is displayed.

---

3. Prediction Lifecycle

3.1 Initial prediction

The first prediction must be generated immediately after the student completes the full 21-question diagnostic.

The diagnostic must produce two independent outputs:

1. Seven initial category Ability Scores, as defined by PRD-014.
2. One initial Predicted SAT Score Range, as defined by this PRD.

The diagnostic prediction must not be calculated from the Ability Scores.

It must use the student's exact Easy, Medium, and Hard result pattern in each category.

3.2 Later predictions

After the diagnostic, predictions must update only when a complete 21-question adaptive practice set is submitted.

Ability Scores continue to update after every finalized answer, but the visible predicted range must not change during an Active Practice Set.

After set completion:

1. Read all seven current Ability Scores.
2. Calculate Reading and Writing Ability.
3. Calculate Math Ability.
4. Calculate Overall Ability.
5. Map Overall Ability to a predicted SAT range.
6. Create an immutable prediction-history entry.
7. Update the Dashboard and Progress page.

3.3 Existing active sets

Prediction updates must not modify:

- The current Active Practice Set
- Its category allocation
- Its difficulty blueprint
- Its selected questions

The updated prediction affects display only.

---

4. Category Weighting

4.1 Reading and Writing Ability

[
RW =
0.45R +
0.45G +
0.10V
]

Where:

- (R) = Reading Comprehension Ability
- (G) = Grammar Ability
- (V) = Vocabulary Ability

4.2 Math Ability

[
M =
0.35A +
0.35AM +
0.15P +
0.15GT
]

Where:

- (A) = Algebra Ability
- (AM) = Advanced Math Ability
- (P) = Problem Solving & Data Analysis Ability
- (GT) = Geometry & Trigonometry Ability

4.3 Overall Ability

[
O=\frac{RW+M}{2}
]

Where (O) is the student’s Overall Ability from `0.0` through `100.0`.

All calculations must use the full decimal Ability Scores.

Ability Scores must not be rounded before weighting.

---

5. Diagnostic-Specific Prediction

5.1 Category diagnostic value

For each of the seven categories, use the exact diagnostic result pattern to assign a category diagnostic score value.

Easy	Medium	Hard	Diagnostic score value
Incorrect	Incorrect	Incorrect	400
Correct	Incorrect	Incorrect	1000
Incorrect	Correct	Incorrect	800
Correct	Correct	Incorrect	1300
Incorrect	Incorrect	Correct	600
Correct	Incorrect	Correct	1200
Incorrect	Correct	Correct	1000
Correct	Correct	Correct	1600

This table is separate from the PRD-014 diagnostic Ability initialization table.

The same response pattern may produce:

- One value for adaptive Ability initialization
- A different value for initial SAT prediction

This distinction is intentional.

5.2 Diagnostic Reading and Writing value

[
RW_D =
0.45R_D +
0.45G_D +
0.10V_D
]

Where each variable is the diagnostic score value assigned from the table in Section 5.1.

5.3 Diagnostic Math value

[
M_D =
0.35A_D +

0.35AM_D +
0.15P_D +
0.15GT_D
]

5.4 Total diagnostic estimate

[
D=\frac{RW_D+M_D}{2}
]

Where (D) is the internal diagnostic estimate from `400.0` through `1600.0`.

The displayed initial prediction is the SAT range containing (D).

Range mapping

Internal diagnostic estimate	Displayed range
1530–1600	1530–1600
1450–1529.999…	1450–1520
1370–1449.999…	1370–1440
1290–1369.999…	1290–1360
1210–1289.999…	1210–1280
1130–1209.999…	1130–1200
1050–1129.999…	1050–1120
970–1049.999…	970–1040
890–969.999…	890–960
810–889.999…	810–880
730–809.999…	730–800
650–729.999…	650–720
570–649.999…	570–640
490–569.999…	490–560

400–489.999…	400–480

The diagnostic estimate becomes the student's immutable starting prediction.

---

6. Post-Diagnostic Ability Mapping

After every completed adaptive set, map Overall Ability to the following score range:

Overall Ability	Predicted SAT range
89.0–100.0	1530–1600
82.0–88.999…	1450–1520
70.0–81.999…	1370–1440
61.0–69.999…	1290–1360
53.0–60.999…	1210–1280
47.0–52.999…	1130–1200
43.0–46.999…	1050–1120
39.0–42.999…	970–1040
36.0–38.999…	890–960
33.0–35.999…	810–880
30.0–32.999…	730–800
26.0–29.999…	650–720
16.0–25.999…	570–640
10.0–15.999…	490–560
0.0–9.999…	400–480

This mapping is intentionally nonlinear.

The Ability Score was designed for adaptive question selection, not as a direct percentage or linear SAT score.

All thresholds must remain centrally configurable.

---

7. Approximate Improvement

The student's improvement must be measured relative to their initial diagnostic prediction.

Each score range has a representative midpoint:

Predicted range	Representative midpoint
1530–1600	1565
1450–1520	1485
1370–1440	1405
1290–1360	1325
1210–1280	1245
1130–1200	1165
1050–1120	1085
970–1040	1005
890–960	925
810–880	845
730–800	765
650–720	685
570–640	605
490–560	525
400–480	440

Calculate:

[

\text{Approximate Improvement}

\text{Current Range Midpoint}

\text{Initial Range Midpoint}
]

Examples:

Initial range: 1130–1200
Current range: 1290–1360
Approximate improvement: +160 points

Initial range: 1370–1440
Current range: 1290–1360
Approximate improvement: −80 points

Display wording:

Approximately +160 points

Approximately −80 points

If the student remains in the same range, approximate point improvement remains unchanged.

Movement inside the range is represented separately through within-range progress.

---

8. Progress Within a Range

The Progress page must show the student’s position within their current predicted range.

The exact internal Ability Score must not be displayed.

8.1 Post-adaptive calculation

For post-diagnostic predictions:

[
Progress =

\frac{O-L}{U-L}
]

Where:

- (O) = Overall Ability
- (L) = lower Ability threshold for the current range
- (U) = upper Ability threshold for the current range

Clamp the result between `0.0` and `1.0`.

Example:

The `1370–1440` range covers Ability `70.0` through below `82.0`.

For Overall Ability `76.0`:

[
Progress=\frac{76-70}{82-70}=0.50
]

The progress indicator displays the student halfway toward the next range.

8.2 Initial diagnostic calculation

For the initial diagnostic prediction, calculate progress using the internal diagnostic estimate's position inside the displayed SAT range.

The student must not see:

- The diagnostic estimate
- The Overall Ability
- Exact threshold values
- Decimal calculations

The interface should communicate progress toward the next predicted range without claiming an exact score.

For the highest range, the indicator may show full completion or "Top predicted range."

---

9. Display Requirements

9.1 Dashboard

The Student Home Dashboard must display:

- `Predicted SAT Score`
- Current predicted total range
- Approximate improvement since the diagnostic

Example:

Predicted SAT Score
1290–1360

Approximately +160 points

The range is the primary metric.

9.2 Progress page

The Progress page must display:

- Current total predicted SAT range
- Approximate improvement since the diagnostic
- Within-range progress
- Historical range changes over completed sets

The Progress page must not display predicted section ranges.

9.3 Estimate disclaimer

An information tooltip must state:

> An estimate based on your PrepHub performance. Your actual SAT score may vary.

The disclaimer should not appear as a large persistent warning.

9.4 Hidden values

Students must not see:

- Category Ability Scores
- Overall Ability
- Diagnostic score values
- Exact internal predicted scores
- Decimal values

- Score-engine thresholds

---

10. Prediction History

Create one immutable Prediction History entry:

- After diagnostic completion
- After every completed adaptive practice set

A history entry must be created even if the displayed range does not change.

Required fields:

predictionId
studentId
sourceType
sourceSetId
readingWritingAbility
mathAbility
overallAbility
internalDiagnosticEstimate
displayedRangeIndex
displayedRangeMinimum
displayedRangeMaximum
representativeMidpoint
approximateImprovement
withinRangeProgress
scoringEngineVersion
createdAt

Source type

DIAGNOSTIC
ADAPTIVE_SET

For a diagnostic prediction:

- `internalDiagnosticEstimate` must be populated.
- Ability values may also be stored for analytics.
- The diagnostic-specific table determines the displayed range.

For an adaptive-set prediction:

- Weighted Ability values must be populated.
- `internalDiagnosticEstimate` must be null.
- The Overall Ability table determines the displayed range.

Historical entries must never be recalculated after configuration changes.

---

11. Configuration and Versioning

The following must be centrally configurable:

- Diagnostic pattern score values
- Reading and Writing category weights
- Math category weights
- Overall Ability thresholds
- Displayed SAT ranges
- Representative range midpoints
- Scoring-engine version

Configuration validation must confirm:

- Reading and Writing weights sum to `1.0`.
- Math weights sum to `1.0`.
- Score ranges cover `400–1600` without overlap.
- Ability thresholds cover `0.0–100.0` without gaps or overlap.
- Range ordering is ascending.
- Every diagnostic pattern has exactly one score value.
- Diagnostic values remain between `400` and `1600`.

Configuration changes must affect only future prediction entries.

Existing prediction history must remain unchanged.

---

12. Error Handling

Do not generate a prediction when:

- The diagnostic is incomplete
- Any diagnostic category lacks exactly one Easy, Medium, and Hard result

- One or more required Category States are missing
- An Ability Score is outside `0.0–100.0`
- Scoring configuration is invalid
- A completed adaptive set has not finalized all 21 answers

When prediction generation fails:

- Preserve the previous valid prediction.
- Log the failure and affected student.
- Do not display a fabricated estimate.
- Show a safe unavailable state if no previous prediction exists.

A failed prediction calculation must not prevent the completed practice set from remaining completed.

---

13. Prediction Pseudocode

13.1 Diagnostic prediction

```
function generateDiagnosticPrediction(student):

  values = {}

  for category in ALL_CATEGORIES:
    results = getDiagnosticResults(student, category)

    assert results contains:
      one Easy result
      one Medium result
      one Hard result

    values[category] =
      diagnosticValueFor(results)

  readingWriting =
    0.45 * values.READING_COMPREHENSION +
    0.45 * values.GRAMMAR +
    0.10 * values.VOCABULARY

  math =
    0.35 * values.ALGEBRA +
    0.35 * values.ADVANCED_MATH +
    0.15 * values.PROBLEM_SOLVING +
```

```
    0.15 * values.GEOMETRY_TRIGONOMETRY

  diagnosticEstimate =
    (readingWriting + math) / 2

  range =
    rangeContainingScore(diagnosticEstimate)

  progress =
    positionInsideScoreRange(
      diagnosticEstimate,
      range
    )

  persist prediction as DIAGNOSTIC

  return range
```

13.2 Adaptive prediction

```
function generateAdaptivePrediction(student, completedSet):

  abilities = loadSevenCategoryAbilities(student)

  readingWriting =
    0.45 * abilities.READING_COMPREHENSION +
    0.45 * abilities.GRAMMAR +
    0.10 * abilities.VOCABULARY

  math =
    0.35 * abilities.ALGEBRA +
    0.35 * abilities.ADVANCED_MATH +
    0.15 * abilities.PROBLEM_SOLVING +
    0.15 * abilities.GEOMETRY_TRIGONOMETRY

  overallAbility =
    (readingWriting + math) / 2

  range =
    rangeForOverallAbility(overallAbility)

  progress =
    positionInsideAbilityThreshold(
      overallAbility,
```

```
    range
  )

initialPrediction =
  loadDiagnosticPrediction(student)

approximateImprovement =
  range.midpoint -
  initialPrediction.range.midpoint

persist prediction as ADAPTIVE_SET

return range
```

---

14. Core Invariants

The following must always remain true:

- The first prediction appears immediately after the completed diagnostic.
- The initial prediction uses exact diagnostic response patterns.
- The initial prediction does not use the Ability-to-range table.
- Later predictions use the seven current Ability Scores.
- Predictions update only after completed sets.
- The student sees only a total SAT range.
- No exact predicted SAT score is displayed.
- Reading and Writing and Math predictions are not displayed separately.
- The diagnostic prediction remains the improvement baseline.
- Score decreases are shown.
- Historical predictions are immutable.
- Configuration changes do not rewrite prediction history.
- Missing or invalid data never produces a fabricated prediction.

---

15. Acceptance Criteria

PRD-016 is correctly implemented when:

1. Completing the diagnostic creates seven Ability Scores and one initial prediction.
2. The initial prediction uses the diagnostic-specific table.
3. Every category pattern maps to the configured diagnostic value.

4. The diagnostic category values are weighted correctly.
5. The initial internal estimate maps to the correct displayed range.
6. No prediction appears before diagnostic completion.
7. Ability changes during an Active Practice Set do not change the visible prediction.
8. Completing an adaptive set creates a new prediction.
9. Later predictions use the nonlinear Overall Ability table.
10. Reading and Writing Ability uses `45% / 45% / 10%`.
11. Math Ability uses `35% / 35% / 15% / 15%`.
12. The student sees only the total SAT range.
13. Approximate improvement uses the initial diagnostic range as its baseline.
14. Score increases and decreases are both displayed.
15. Within-range progress changes when Overall Ability changes inside the same range.
16. A prediction-history entry is created after every completed set.
17. A history entry is created even when the displayed range does not change.
18. Historical predictions remain unchanged after configuration updates.
19. Invalid or missing adaptive state does not generate a prediction.
20. Dashboard and Progress use the same latest stored prediction.

---

16. Out of Scope for V1

The following are not included:

- Exact predicted SAT scores
- Predicted section-score ranges
- Student-visible Ability Scores
- Score prediction during an Active Practice Set
- Different scoring models by school or student
- Automatic calibration using official SAT outcomes
- Verified-score collection
- Machine-learned score prediction
- Confidence intervals beyond the displayed score ranges
- Percentile prediction
- National score comparisons

End of PRD-016